IaaS- Infrastructure as a Service :

IaaS (Infrastructure as a Service) is a cloud computing model where a provider delivers basic computing infrastructure—like servers, storage, and networking—over the internet. Instead of buying and maintaining physical hardware, you rent these resources on demand.

Simple idea

Think of IaaS as **renting a virtual data center**. You control the systems and applications, but the provider manages the physical hardware.

Key features

- **Virtual machines (VMs):** You can create and run servers in the cloud
- **Storage:** Save data without managing physical disks
- **Networking:** Configure IPs, firewalls, and load balancers
- **Scalability:** Increase or decrease resources anytime
- **Pay-as-you-go:** Only pay for what you use

What you manage vs. provider manages

You manage	Provider manages
Applications	Physical servers
Data	Data centers
Operating system	Networking hardware
Runtime & middleware	Virtualization infrastructure

Examples of IaaS providers

- Amazon Web Services (EC2)
- Microsoft Azure (Virtual Machines)
- Google Cloud Platform (Compute Engine)

When to use IaaS

- Hosting websites or apps
- Running development and testing environments
- Big data processing
- Backup and disaster recovery

Example: E-commerce Website using IaaS

Imagine you are building an online shopping site like a small version of **Amazon**.

🔧 Step-by-step using IaaS

1. Rent infrastructure (instead of buying servers)

You go to a cloud provider like:

- Amazon Web Services (EC2)
- Microsoft Azure (VMs)

You create a **virtual server (VM)** in minutes.

PaaS (Platform as a Service)

PaaS is a cloud model where the provider gives you a **ready-made platform** to build, run, and deploy applications—without worrying about servers, OS, or infrastructure.

Key features

- **No server management**
- **Built-in development tools**
- **Auto scaling**
- **Faster deployment**
- **Developer-friendly**

When to use PaaS

- Building web or mobile apps
- Rapid development (startups, hackathons)
- API development
- Team collaboration projects

What you manage vs provider manages

You manage	Provider manages
Application code	Servers & hardware
Data	Operating system
App configuration	Runtime (Java, Python, etc.)
	Scaling & infrastructure

Popular PaaS platforms

- Google App Engine
- Microsoft Azure App Service
- Heroku

Real-world example (same e-commerce idea)

Imagine you're building an online store

With PaaS:

1. Go to Heroku or Microsoft Azure App Service
2. Upload your code (Java / Node.js / Python)
3. Click **Deploy**
4. Your app is live!

No need to:

- Install OS
- Configure servers
- Manage scaling

SaaS (Software as a Service)

SaaS is a cloud model where you use **ready-made software over the internet**—no installation, no setup, no maintenance.

Key features

- No installation required
- Accessible anywhere (browser/mobile)
- Automatic updates
- Subscription-based (monthly/yearly)
- Easy to use

When to use SaaS

- Email services
- Video streaming
- Online document editing
- CRM tools (like sales software)

What you manage vs provider manages

You manage	Provider manages
Your data	Application
Basic settings	Servers & infrastructure
	OS, updates, security, scaling

Popular SaaS applications

- Google Docs
- Microsoft 365
- Gmail
- Netflix

Real-world example

Example: Using email

Instead of installing software:

1. Open Gmail in browser
2. Login
3. Send/receive emails instantly

You don't care about:

- Servers
- Storage systems
- Updates

Everything is handled by Google.

Detailed comparison

Feature	IaaS	PaaS	SaaS
Full form	Infrastructure as a Service	Platform as a Service	Software as a Service
Control level	High	Medium	Low
User manages	OS, apps, data	Apps & data	Only usage
Provider manages	Hardware, networking	OS, runtime, infra	Everything
Setup time	More	Moderate	Very fast
Flexibility	Very high	Limited	Very limited
Technical skill	High	Medium	Low
Use case	Full system control	App development	Daily usage apps